

Advanced 3D Deep Transfer Learning For Multistage Alzheimer's Disease Diagnosis

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Abstract

An advanced 3D deep transfer learning for multistage AD diagnosis was developed with:

- high precision/sensitivity (93%) in predicting mild cognitive impairment (MCI) and AD altogether combined.
- excellent recall sensitivity (94%) in predicting MCI with mitigated false negative rate comparing to other published methods.

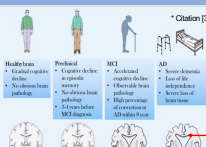
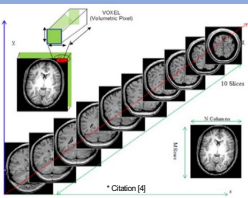
The innovative process involves:

- smRI image preprocessing (1a) and a U-Net semantic segmentation (1b) to identify the crop-sized region of interest (ROI) containing the largest lateral ventricle volume, facilitating the model to focus on relevant information, reduced training time and computational cost.
- the pretrained weights on 2D ImageNet of more than 1 million natural images from public repository^[1] transferring into a 3D convolutional neural network (CNN) of RGB dataset of smRI ROI.
- a modified deep transfer learning architecture VGG16 with additional normalizations, dropouts, and training hyperparameters.

Background

Brain atrophy, as measured on smRI, is a powerful biomarker of AD pathology. However, manual interpretation of these images can be challenging and time-consuming for radiologists due to subjectivity and precision issues.^[2]

smRI scans contain many slices as 2D images stacked together to form a 3D volume.^[4]



Enlarged Lateral Ventricle
Shrunk Hippocampus

Dataset and Tools

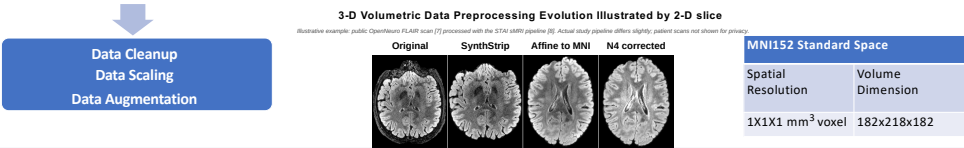
-T1-weighted Magnetization-Prepared Rapid Acquisition Gradient Echo axial structural MRI scans was obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI).^[5]
-The total of 964 sample cases were filtered from ADNI/1, ADNI/GO, ADNI/2, and ADNI/3.

- FSL
- SimpleITK
- Niabel
- Keras
- Numpy
- Pandas
- Scikit Learn
- GitHub
- Classification-Models-3D

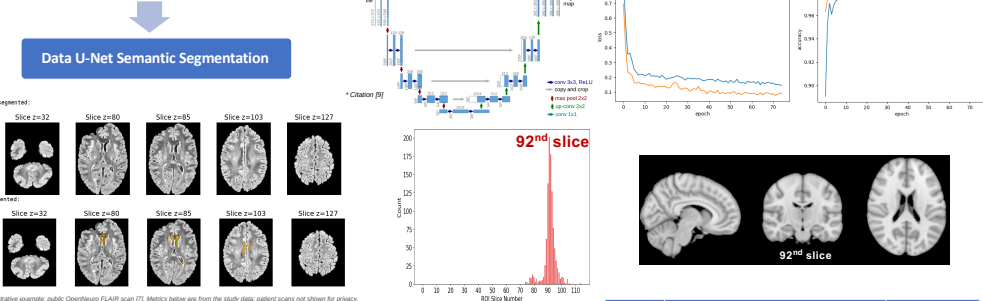
Challenges and Approaches

1a. The volumetric data from smRI is significantly large, with various biases due to differences in scanners, parameters, inherent random noise, and intensity inhomogeneity.

True Labels for Data Distribution			
CN (Cognitive Normal)	MCI (Mild Cognitive Impairment)	AD (Alzheimer's Disease)	Total
329	584	150	964



1b. Deeper neural networks for machine learning require high computational costs and longer training time.



- U-Net trained on manually labeled slices from different regions of multiple samples' volumes reaching a testing dice loss of 0.0884 and testing accuracy of 0.9895
- Out of the 182 slices from each sample, the crop-sized region of interest (ROI) was identified as containing the largest lateral ventricle volume with 20 slices in total and centered at the 92nd slice.

	CN	MCI	AD	Percentage (Number)
Training	0.26568099	0.14937429	0.58494471	80%
Validation	0.2605905	0.14377407	0.59563543	10%
Test	0.27017745	0.15569548	0.57412707	10%

3. VGG16 with its simplicity and effectiveness, is a widely used deep learning model. However, it still needs further modifications to mitigate overfitting, avoid vanishing gradient problem, and optimize computational efficiency.

VGG16-Keras 3D Convolutional Neural Network Training with additional layers of batch normalizations / dropout , and hyperparameter modifications

Multisets Performance Metrics

Performance Metrics	Training Set	Validation Set	Testing Set
Loss	0.0348	0.3002	1.6195
Accuracy	0.9965	0.8906	0.8438
AUC	0.9997	0.9732	0.9129
Precision	0.9965	0.9040	0.8629
Recall	0.9942	0.8828	0.8359

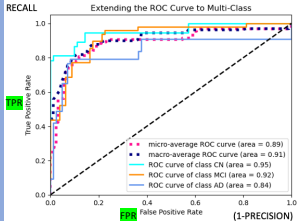
Model Prediction Evaluations

Multiclass Confusion Matrix Analysis

		Predicted Label (CN, MCI, AD)				
Cognitive Normal (CN)	True Label	30	5	2	0.81	Recall (Sensitivity)
		0.97	0.71	0.86	0.82	
Mild Cognitive Impairment (MCI)	True Label	1	45	2	0.94	Precision
		0.92	0.93	0.88	0.88	
Alzheimer's Disease (AD)	True Label	5	13	25	0.58	Precision
		0.92	0.93	0.88	0.88	

- Overall high precision/sensitivity in predicting MCI and AD altogether combined (~93%)
- Excellent recall sensitivity (94%) in predicting MCI, which is clinically crucial for accurate diagnosis of early-onset AD with a mitigated false negative rate

Multiclass One-vs-All AUC-ROC



- Overall, excellent prediction and differentiation for CN and MCI

Methods Comparisons^[6]

Author (Year)	Training Size (of Images)	Segmentation	Pre-training	CNN Variant	Classification	Accuracy	Recall Sensitivity (MCI vs CN)	One-All ROC (CN, AD, MCI)
Current work (2024)	ADNI 15,424 images	Slice with largest lateral ventricle volume	Interdomain 2D ImageNet natural image	ResNet-18 3D ResNet-34 3D ResNet-50 3D VGG16 3D	Multi	0.45 0.53 0.47 0.82	0.94, 0.98	0.95, 0.84, 0.92
Raza N. (2023)	ADNI 4,012 images	Slices with gray matter	Intradomain 2D GM images	DenseNet-169 3D	Multi	0.60	-	-
Majumdar, M. (2019)	OASIS 382 images	Interdomain 2D ImageNet natural image	Interdomain 2D ImageNet natural image	AlexNet 3D	Multi	0.60	-	-
Khan N.M. (2019)	ADNI 15,424 images	Slice with largest lateral ventricle volume	Interdomain 2D ImageNet natural image	VGG16 3D	Multi	0.95	-	-
Hou, M. (2017)	OASIS 382 images	Interdomain 2D ImageNet natural image	Interdomain 2D ImageNet natural image	VGG16 3D	Binary	0.92	-	-
Liu et al (2020)	ADNI 15,424 images	Random cropping	Intradomain age set	ResNet-18 3D AlexNet 3D	Multi	0.68	-	0.90, 0.88, 0.65
Payan A. (2015)	ADNI 117,708 images	Random 3D patch	Interdomain 2D ImageNet natural image	Sparse Autoencoder + 3D CNN	Multi	0.89	-	-
Gupta A. (2013)	ADNI 103,683 images	ZCA whitened natural images	Interdomain 2D ImageNet natural image	Sparse Autoencoder + 3D CNN	Multi	0.85	0.74, 0.92	-

- The current method innovates in components of training size, segmentation, pre-training method and CNN architecture with high sensitivity to differentiate MCI from others (94%) and MCI from CN (98%)

Future Work

- Test other image datasets: OASIS, AIBL etc
- Launch Pilot Website for Virtual Diagnosis Assistant <https://sites.google.com/view/rapid-diagnosis-assistant/home>

*ADNI data and components of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usm.edu). As such, the investigators within the ADNI contributed to the design and implementation of ADNI and/or provided data that did not participate in analysis or writing of this report. A complete listing of ADNI investigators can be found at: http://adni.loni.usm.edu/wp-content/uploads/how_to_use/ADNI/acknowledgement.pdf. Data used in the preparation of this article were obtained from the Alzheimer's Disease Neuroimaging Initiative (ADNI) database (adni.loni.usm.edu). The ADNI was launched in 2003 as a public-private partnership led by Principal Investigator Michael W. Weiner, MD. The primary goal of ADNI has been to develop and test standard imaging and cognitive assessment protocols, and to establish a large database of standard imaging and cognitive assessment protocols. The ADNI database is distributed by the Laboratory for Neuro Imaging at the University of Southern California. ADNI data are disseminated by the Laboratory for Neuro Imaging at the University of Southern California.